

Organic Chemistry Practical: Preparative Organic Synthesis & Laboratory Techniques

10 Most Frequently Asked Practical Examination Questions (Past 4 Years Analysis: 2021–2024)

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 — Advanced Practical Skills (Past 10 Years Archive)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours 30 Minutes	TOTAL MARKS	100 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm³; balance readings to 0.01 g / 0.001 g.
 - Thermometer readings to the nearest 0.5 °C.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: The Principles, Thermodynamics, and Apparatus Setup of Heating Under Reflux

Cambridge Syllabus Reference: 9701/35/M/J/22/Q2 · [10 Marks]

In organic synthesis, many reactions between organic molecules have high activation energies and proceed very slowly at room temperature. Heating under reflux allows prolonged boiling without loss of material.

(a) Define 'reflux' in laboratory organic chemistry. [2]

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(b) Explain two advantages of reflux over heating in an open beaker or conical flask. [3]

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(c) State why a stopper must NEVER be placed in the top of a reflux condenser. [3]

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(d) Explain why electric heating mantles or water baths are preferred over Bunsen burners when refluxing organic liquids. [2]

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Question 2: Physics of Boiling: Role and Mechanism of Anti-Bumping Granules

Cambridge Syllabus Reference: 9701/31/O/N/21/Q2 · [10 Marks]

Anti-bumping granules (small pieces of unglazed porcelain or carborundum) are added to distillation and reflux flasks before heating.

(a) Explain the physical mechanism of how anti-bumping granules promote smooth boiling. [4]

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(b) Describe what happens if a liquid is heated without anti-bumping granules (the phenomenon of 'bumping'). [3]

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(c) Explain why anti-bumping granules must NEVER be added to a hot liquid near its boiling point. [3]

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Question 3: Heat Exchanger Design: Fluid Dynamics of Liebig Condenser Water Jackets

Cambridge Syllabus Reference: 9701/33/O/N/22/Q2 · [10 Marks]

In standard distillation and reflux setups, cold water from the tap must be connected to the condenser in a specific direction.

(a) Specify which nozzle of a Liebig condenser should be connected to the cold water inlet and which to the sink drain. [2]

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(b) Explain the physical consequences if water is erroneously fed into the top nozzle. [4]

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(c) Explain why countercurrent water flow in distillation provides superior thermal efficiency. [4]

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Question 4: Liquid-Liquid Extraction: Principles and Safety of Separating Funnel Procedures

Cambridge Syllabus Reference: 9701/34/M/J/23/Q2 · [10 Marks]

A crude preparation of 1-bromobutane (density = 1.28 g cm^{-3}) is washed in a separating funnel with aqueous sodium hydrogencarbonate (density $\sim 1.05 \text{ g cm}^{-3}$).

(a) State whether the 1-bromobutane layer is the upper or lower layer. Justify your answer. [2]

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(b) Describe the exact procedure for shaking and venting the separating funnel during washing with aqueous NaHCO_3 . [4]

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(c) Identify the gas responsible for pressure buildup and write an equation for its production. [2]

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(d) Explain why the glass stopper at the top must be removed before draining liquid through the bottom stopcock. [2]

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Question 5: Selection, Mechanism, and Removal of Anhydrous Chemical Drying Agents

Cambridge Syllabus Reference: 9701/32/O/N/23/Q2 · [10 Marks]

After extraction in a separating funnel, the organic liquid remains cloudy due to microscopic suspended droplets of water. Anhydrous sodium sulfate, Na_2SO_4 , is added.

(a) State how anhydrous sodium sulfate removes water from the organic liquid. [2]

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(b) State two visual indications that confirm the organic liquid is completely dry. [3]

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(c) Name two alternative anhydrous drying agents commonly used in organic chemistry. [2]

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(d) Describe how the drying agent is separated from the dry liquid prior to distillation. [3]

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Question 6: Kinetic vs Thermodynamic Control in the Oxidation of Ethanol: Ethanal vs Ethanoic Acid

Cambridge Syllabus Reference: 9701/33/M/J/21/Q2 · [10 Marks]

The oxidation of ethanol can be controlled to produce either ethanal (b.p. 21°C) or ethanoic acid (b.p. 118°C).

- Method A: Dropwise addition of ethanol to acidified dichromate in a distillation apparatus.
- Method B: Heating ethanol with excess acidified dichromate under reflux for 30 minutes.

(a) State which product is synthesized in Method A and which in Method B. [2]

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(b) Explain why immediate distillation in Method A stops the oxidation at ethanal. [3]

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(c) Write the balanced oxidation equations for both conversions using [O] as oxidizing agent. [3]

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(d) Explain why excess potassium dichromate is required in Method B. [2]

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Question 7: Vapor Condensation Engineering: The Use of Ice–Water Receiver Baths**Cambridge Syllabus Reference: 9701/35/M/J/24/Q2 · [10 Marks]**

During the synthesis and distillation of volatile organic compounds with low boiling points (such as ethanal, b.p. 21°C, or propanone, b.p. 56°C), the receiving conical flask or collection vial is placed inside a beaker filled with an ice–water slurry.

(a) Explain why receiver cooling is essential when collecting ethanal. [3]

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(b) State two hazards that arise if volatile organic vapors escape into the laboratory. [3]

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(c) Explain why an ice–water slurry is colder and more effective than dry ice blocks or ice cubes alone. [2]

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(d) Suggest an additional precaution involving the receiver adapter to minimize vapor loss. [2]

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Question 8: Stoichiometric Yield Analysis and Limiting Reactant Determinations in Organic Preparations

Cambridge Syllabus Reference: 9701/34/M/J/24/Q2 · [10 Marks]

A student prepares ethyl ethanoate by reacting 9.20 g of ethanol ($M_r = 46.0$) with 15.00 g of pure ethanoic acid ($M_r = 60.0$) in the presence of concentrated sulfuric acid catalyst. After purification, 11.44 g of pure ethyl ethanoate ($M_r = 88.0$) is obtained.

(a) Calculate the moles of ethanol and ethanoic acid used, and identify the limiting reactant. [3]

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(b) Calculate the theoretical yield of ethyl ethanoate in grams. [2]

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(c) Calculate the percentage yield of ethyl ethanoate obtained. [2]

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(d) Explain why esterification reactions never achieve 100% theoretical yield even with perfect laboratory technique. [3]

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Question 9: Systematic Analysis of Practical and Chemical Yield Loss in Multistep Preparations

Cambridge Syllabus Reference: 9701/31/M/J/24/Q2 · [10 Marks]

In the preparation of 1-bromobutane from butan-1-ol, the typical student yield ranges from 60% to 75%. Candidates are evaluated on identifying where the missing 25% to 40% of material is lost.

(a) Identify two distinct chemical side reactions that reduce the yield of 1-bromobutane. [4]

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(b) Identify two physical/mechanical avenues of product loss during the purification procedure. [4]

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(c) Explain why washing with concentrated hydrochloric acid removes unreacted butan-1-ol. [2]

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Question 10: Thermometric Characterization: Determining the Purity of Liquid Organic Products

Cambridge Syllabus Reference: 9701/32/O/N/24/Q2 · [10 Marks]

The purity of a synthesized liquid product is checked by monitoring the vapor temperature during final redistillation, or by using a micro-boiling tube method with an inverted capillary tube.

(a) Describe how the thermometer bulb must be positioned in a distillation still-head for accurate boiling point measurement. [3]

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(b) Contrast the distillation temperature profile of a pure substance with that of an impure sample. [4]

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(c) In Siwoloboff's micro-method, explain what observation signifies that the true boiling point has been reached. [3]

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Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Organic Chemistry Practical: Preparative Organic Synthesis & Laboratory Techniques

Question 1: The Principles, Thermodynamics, and Apparatus Setup of Heating Under Reflux [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Continuous boiling of a liquid reaction mixture such that vapors condense on a vertical condenser and drip back into the reaction vessel [2].	[2]
(b)	1. Prevents loss of volatile reactants, solvent, and products by evaporation [1.5]; 2. Maintains a constant, elevated reaction temperature equal to the boiling point of the mixture [1.5].	[3]
(c)	Creates a completely sealed, closed system [1]; heating causes gas expansion and vapor pressure to build up rapidly, causing an explosion of the glassware [2].	[3]
(d)	Organic liquids and vapors are highly flammable; eliminates open flames that could ignite flammable vapors [2].	[2]

Question 2: Physics of Boiling: Role and Mechanism of Anti-Bumping Granules [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Granules have microscopic pores and rough surfaces containing trapped micro-pockets of air [2]; these act as nucleation sites where vapor bubbles form steadily and release smoothly [2].	[4]
(b)	The liquid can become superheated above its normal boiling point without boiling [1.5]; sudden nucleation triggers an explosive, violent surge of large vapor bubbles that ejects hot liquid out of the flask ('bumping') [1.5].	[3]
(c)	Adding granules to superheated liquid introduces countless nucleation sites instantly, causing instantaneous explosive flash boiling that erupts hot liquid over the user [3].	[3]

Question 3: Heat Exchanger Design: Fluid Dynamics of Liebig Condenser Water Jackets [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Water inlet: bottom / lower nozzle [1]; Water outlet: top / upper nozzle to drain [1].	[2]
(b)	Water will trickle down under gravity leaving a large air pocket / void at the top of the jacket [2]; the condenser jacket will not fill completely, resulting in poor heat transfer, incomplete condensation, and escape of flammable vapors [2].	[4]
(c)	In distillation, countercurrent flow ensures that the coolest water meets the exiting vapor at the lowest point, maximizing the temperature gradient along the entire length of the tube [2]; this guarantees complete condensation before vapors can exit into the collection adapter [2].	[4]

Question 4: Liquid-Liquid Extraction: Principles and Safety of Separating Funnel Procedures [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Lower layer [1]; 1-bromobutane has a higher density (1.28 g cm^{-3}) than aqueous sodium hydrogencarbonate [1].	[2]
(b)	Hold the stopper firmly with one hand and invert the separating funnel [1]; immediately point the stem away from all people into the back of the fume hood and open the tap/stopcock to vent pressure [1]; close tap, shake gently for 3-5 seconds, invert and vent again [1]; repeat until no further gas 'hiss' is heard, then place in retort ring stand and remove stopper [1].	[4]
(c)	Carbon dioxide, CO_2 [1]; $\text{H}^+(\text{aq}) + \text{HCO}_3^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$ [1].	[2]
(d)	If stopper remains in place, a partial vacuum develops inside the funnel above the liquid as it drains, which arrests liquid flow [2].	[2]

Question 5: Selection, Mechanism, and Removal of Anhydrous Chemical Drying Agents [10 Marks]

Part	Marking Points & Expected Observations	Marks
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(a)	Chemically binds water into its crystal lattice by forming hydrated salt crystals: $\text{Na}_2\text{SO}_4(\text{s}) + 10\text{H}_2\text{O} \rightarrow \text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}(\text{s})$ [2].	[2]
(b)	1. The organic liquid turns from cloudy/turbid to completely clear and transparent [1.5]; 2. The drying agent particles swirl freely like fine snow rather than sticking/clumping together at the bottom of the flask [1.5].	[3]
(c)	Anhydrous calcium chloride (CaCl_2) [1]; anhydrous magnesium sulfate (MgSO_4) [1].	[2]
(d)	Careful gravity filtration through fluted filter paper in a glass funnel into a clean flask, or careful decantation of the clear liquid [3].	[3]

Question 6: Kinetic vs Thermodynamic Control in the Oxidation of Ethanol: Ethanal vs Ethanoic Acid [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Method A: Ethanal, CH_3CHO [1]; Method B: Ethanoic acid, CH_3COOH [1].	[2]
(b)	Ethanal has a very low boiling point (21°C) and vaporizes as soon as it forms [1.5]; it distills out of the reaction flask and into the receiver, removing it from further contact with the oxidizing agent [1.5].	[3]
(c)	Method A: $\text{CH}_3\text{CH}_2\text{OH} + [\text{O}] \rightarrow \text{CH}_3\text{CHO} + \text{H}_2\text{O}$ [1.5]; Method B: $\text{CH}_3\text{CH}_2\text{OH} + 2[\text{O}] \rightarrow \text{CH}_3\text{COOH} + \text{H}_2\text{O}$ [1.5].	[3]
(d)	To ensure complete oxidation of both ethanol and intermediate ethanal into ethanoic acid without unreacted alcohol remaining [2].	[2]

Question 7: Vapor Condensation Engineering: The Use of Ice-Water Receiver Baths [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Ethanal boils at 21°C , which is at or below typical laboratory ambient room temperature [1.5]; without cooling, ethanal would remain as a vapor and evaporate into the air rather than collecting as a liquid [1.5].	[3]
(b)	1. Fire and explosion hazard: organic vapors are highly flammable and can ignite on contact with electrical equipment or flames [1.5]; 2. Health hazard: vapors are toxic/irritants causing dizziness and respiratory irritation [1.5].	[3]
(c)	Water in the slurry provides complete liquid contact with the outer surface of the flask, maximizing surface area for rapid heat conduction [2].	[2]
(d)	Attach an adapter with a vent tube leading into a fume cupboard or submerge receiver neck in a cold trap [2].	[2]

Question 8: Stoichiometric Yield Analysis and Limiting Reactant Determinations in Organic Preparations [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Moles ethanol = $9.20 / 46.0 = 0.200 \text{ mol}$ [1]; Moles ethanoic acid = $15.00 / 60.0 = 0.250 \text{ mol}$ [1]; Ratio is 1:1 $\rightarrow$ Ethanol is the limiting reactant [1].	[3]
(b)	Theoretical moles = 0.200 mol [1]; Theoretical mass = $0.200 \times 88.0 = 17.60 \text{ g}$ [1].	[2]
(c)	% yield = $(11.44 / 17.60) \times 100 = 65.0\%$ [2].	[2]
(d)	Esterification is a reversible, dynamic equilibrium reaction: $\text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH} \rightleftharpoons \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$ [1.5]; the reaction stops when equilibrium is reached before all limiting reactant can be converted into product [1.5].	[3]

Question 9: Systematic Analysis of Practical and Chemical Yield Loss in Multistep Preparations [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	1. Acid-catalyzed elimination / dehydration of butan-1-ol to form gaseous but-1-ene [2]; 2. Intermolecular condensation between two butan-1-ol molecules to form di-n-butyl ether ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{-O-CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$) [2].	[4]
(b)	1. Incomplete separation / slight solubility of 1-bromobutane in aqueous wash layers in the separating funnel [2]; 2. Liquid retained on glassware walls, drying agent residue (absorbed by Na_2SO_4 crystals), or left in distillation flask residue [2].	[4]

(c)	Concentrated HCl protonates the alcohol to form an oxonium ion ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}_2^+$), which is highly polar and dissolves into the aqueous acid layer, separating from the non-polar 1-bromobutane [2].	[2]
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Question 10: Thermometric Characterization: Determining the Purity of Liquid Organic Products [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	The bulb of the thermometer must be positioned exactly level with or just below the entrance to the side-arm condenser [2]; this ensures it is completely bathed in the escaping pure equilibrium vapor entering the condenser [1].	[3]
(b)	Pure substance: Distills at a sharp, constant, fixed temperature equal to the literature boiling point (e.g. within $\pm 0.5^\circ\text{C}$) [2]; Impure sample: Boils over a broad temperature range and the initial boiling point is depressed or elevated depending on the impurity [2].	[4]
(c)	As temperature rises, a rapid continuous stream of bubbles escapes from the inverted capillary [1.5]; upon extinguishing the heat, the exact boiling point is recorded when the stream of bubbles stops and liquid just rushes back up into the capillary [1.5].	[3]